

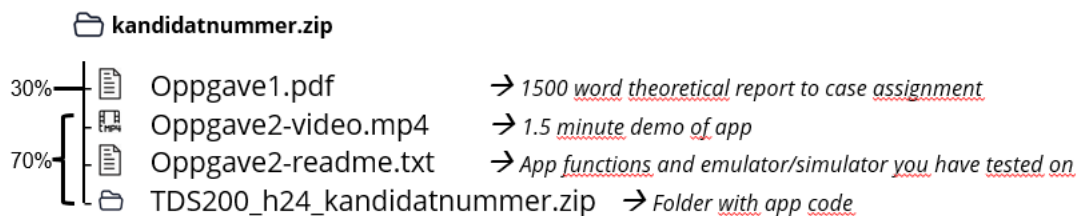
CONTINUATION EXAM

TDS200 Kryssplattform

Allowed aids: All

Autumn 2024

You are required to complete two tasks, and your performance will be evaluated based on both technical and theoretical competence. The following pages contain the details of the two tasks. Your submission should follow the below structure:



Important Information

- Read through the entire exam text before starting.
- Always follow good referencing practice. When using quotes, code snippets, statistics, images, figures, or similar elements that you have not created yourself, it is essential to provide proper citations. Follow the referencing guidelines established by Høyskolen Kristiania.
- Plagiarism must not occur. It is not allowed to copy code without providing a proper reference. Using TDS200 course lecture code is not regarded as plagiarism. However, you should also give citation when you refer to the lecture code.
- Using component library and npm modules is not considered plagiarism, and it is encouraged to demonstrate your competence utilizing npm modules and third-party libraries. Remember to provide appropriate references too.

Task 1 – Theoretical competence (weighting: ~30%)

This assignment requires you to critically evaluate cross-platform development frameworks, examining their characteristics, performance, and applicability in real-world scenarios. You are expected to write a report that consists approximately **1,000 words, excluding the reference list**.

The general requirement is that your report must be written in your own words. Citations should be used when referencing external sources to clearly distinguish between your own work and that of others. Supplementary materials such as code snippets, screenshots, diagrams, or models are encouraged to enhance the comprehensibility of your answers. The report must be submitted in either PDF or Word format. The assigned points for each task serve as an approximate guideline, reflecting the expected level of efforts. To achieve a higher score, it is recommended to blend personal experiences and insights with information from external sources as needed. Answer the following two sub-tasks:

- 1) Discuss the advantages and limitations of a cross-platform approach compared to native development. (~15%)
- 2) In this course, you have been introduced to various cross-platform development frameworks. Compare two types of widely used frameworks, i.e., hybrid (Ionic + Vue) and cross-compiled (React Native) and analyse their performance, user interface, ease of integration with platform-specific features, scalability and maintainability, among other features (~15%)

Task 2 – Technical competence (weighting: ~70%)

For this task, you are going to create an application called PaddleForOcean.

Key evaluation criteria for the exam:

- The solution must be built and run without errors and can be run with Web browsers, as well as Android emulator **OR** iOS simulator.
- Comment the code to illustrate your understanding of code and explain complex logic when necessary.
- The code should adhere to best practices to enhance code quality, including proper naming conventions, code structures and emphasizing code reusability.
- Ensure the app is user-friendly with all functionalities working as expected.
- Any external libraries, modules, or packages required for the project should be integrated appropriately.

Creation and delivery of exam project:

- Create your own Firebase and there must be some data available at Firebase so that the exam sensor can see that the solution works.
- Make a short video (screen recording without sound) of the app showing the functions you have implemented. Limit the video to no more than 1.5 minutes and ensure it highlights any advanced features that may affect the grade.
- Please also attach a file called Task2-readme.txt to describe what functionality you have implemented in the app and whether you have run it on Android emulator, iOS simulator or both.
- **Remove the node_modules** folder before uploading to WISEFLOW!
- Make sure the package.json file contains all npm dependencies, expecting the exam sensor to install npm packages with the command “npm install”
- The project folder you delivered shall be named as “TDS200_h24_your candidate number.zip”

Case: PaddleForOcean App

Technical Task

Please develop the **PaddleForOcean** application using either **Ionic + Vue** or **React Native** for frontend development (in alignment with the technologies taught in class), and **Firebase** as Backend as a Service.

Background

Remember to state your candidate number on your delivery folder.

You work for a non-profit organization dedicated for ocean conservation and cleanup initiatives. Your leader has assigned you a new project: to develop a sustainable web application that recruit paddle enthusiasts to combine their passion for paddling with ocean cleanup. **This innovative digital platform allows users to register themselves, sign up for paddle cleanup trips, create new cleanup mission, track cleanup efforts, etc.** By leveraging this application, the organization aims to attract more participants and strengthen its mission to achieve a cleaner and healthier ocean.

Four primary requirements for implementation (no priority order by numbering):

- **Authentication Page:** Manage user access through registration and login. Only authenticated users can access the application.
- **Overview Page:** This overview page displays a list of available paddle cleanup trips for easy browsing. Each trip include address, picture, date, time, maximum number of participants.
- **Detail Page:** Clicking on a paddle trip opens a detail page with full information. Please include at least below information:
 - trip information such as picture, address, date and time, organizer details, cleanup goal (i.e., 1 kg wastes), max limit of participants.
 - Participant list: names or avatars of users that have signed up.
 - Sign-up button: allows user to sign-up if there is available slot left. If you have already signed up, the button will be disabled. If there is no available slot left, the button will be disabled too.
- **Upload Page:** The upload page allows users to create and upload a new cleanup trip. The new trip will be added to the Overview page upon submission. Required fields include trip title, date, location, image, max participants, cleanup goal, and organizer details.

The four points mentioned above present the minimum requirement for the assignment. For those aiming for a higher grade, more functionalities and complexity in the implementation is expected. Please check below list to see examples of functionalities that can contribute to a higher grade. Note that the app should be designed for the sensor to easily test with minimal manual configuration.

Examples of extended functionalities that can achieve a higher grade:

- Give users the opportunity to filter paddle trips by categories (location, date, maximum number of participants, etc).
- Allow users to search for paddle trips by keywords.
- User profile page with basic details and their signed-up and/or completed trips.
- Provide GPS (Geolocation) information to allocate where the trip will take place.

Remember to state your candidate number on your delivery folder.

- Incorporate a map view, i.e., Google Maps in the app.
- Allow users to mark the trip as upcoming, in progress, or completed.
- An archive page that displays successfully completed trips. It helps to maintain a historical record of past events.
- Allow users to leave feedback on trips (e.g., rating system for organizers).

Appendix: Grading scale

In the exam text, the approximate maximum score per task is shown as below:

Symbol	Fra eksamensavdelingen
A	90-100 poeng
B	80 – 89 poeng
C	60 - 79 poeng
D	50 - 59 poeng
E	40 - 49 poeng
F	0 - 39 poeng

- Task 1 = ~30 points
- Task 2 = ~70 points
- Total = 100 points